THEORY

# Training Program 01

## Advanced NLP & Large Language Models (LLMs)

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**M01: Text Preprocessing**

M02: Word and Sentence Embeddings

M03: Transformer Architectures

M04: Entity-Centric NLP

M05: Applied NLP Tasks

M06: Multimodal NLP

M07: LLM Fundamentals

M08: Prompt Engineering

M09: Fine-Tuning & PEFT

M10: RAG & Real-World Projects

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MODULE M01

# TEXT PREPROCESSING

CHAPTER M01.2

## BPE Byte Pair Encoding

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1. Getting Started with Text Processing
2. Fundamentals of Tokenization
3. Byte-Pair Encoding (BPE) and Subword Tokenization
4. Advanced Subword Tokenization with SentencePiece
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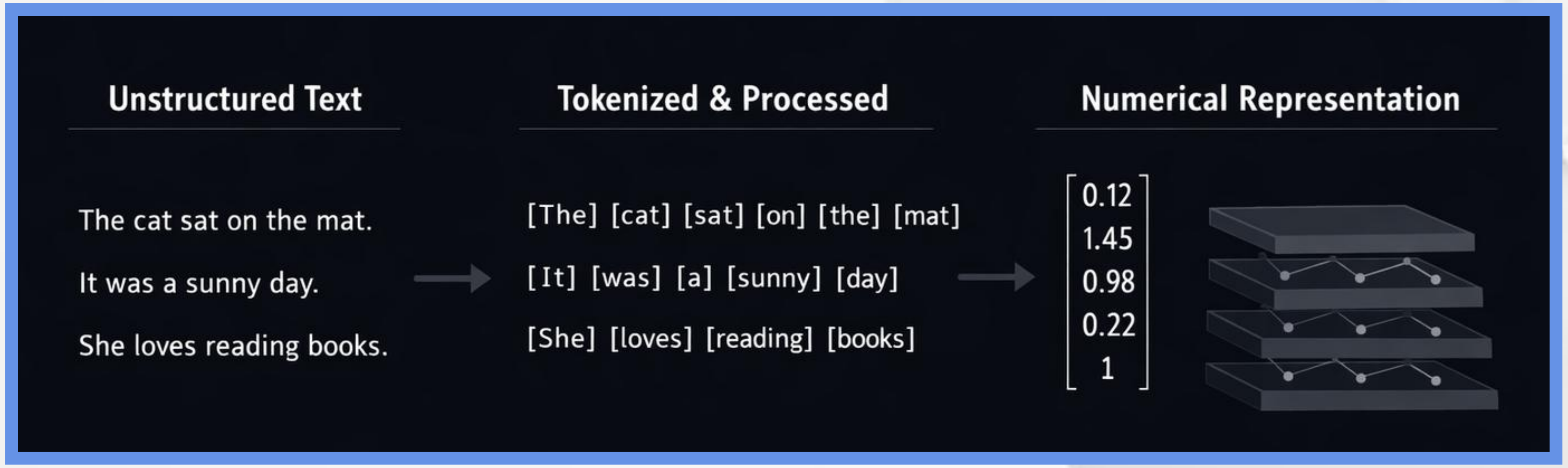
# M01.1 BPE

## Chapter 1.1 Getting Started with Text Processing.

- Text processing is not a preliminary task, but the foundation of all modern NLP systems.
- Before any model can learn, raw human language must be transformed into a structured numerical form.
- This transformation defines what a model can and cannot understand.

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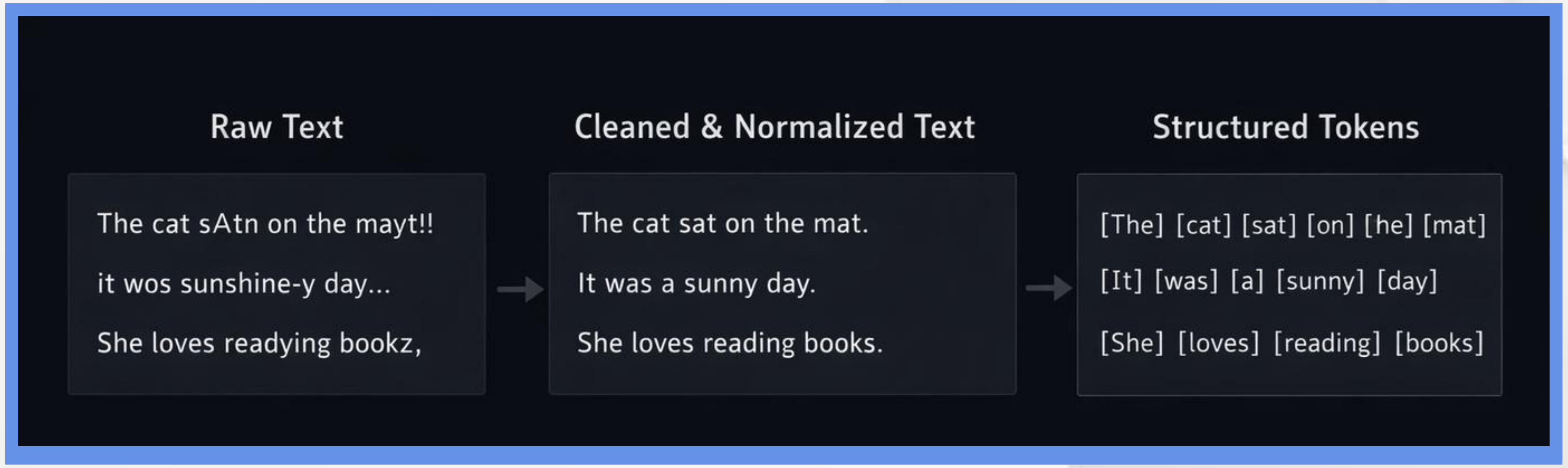
## M01.1 BPE

# Chapter 1.1 Getting Started with Text Processing.

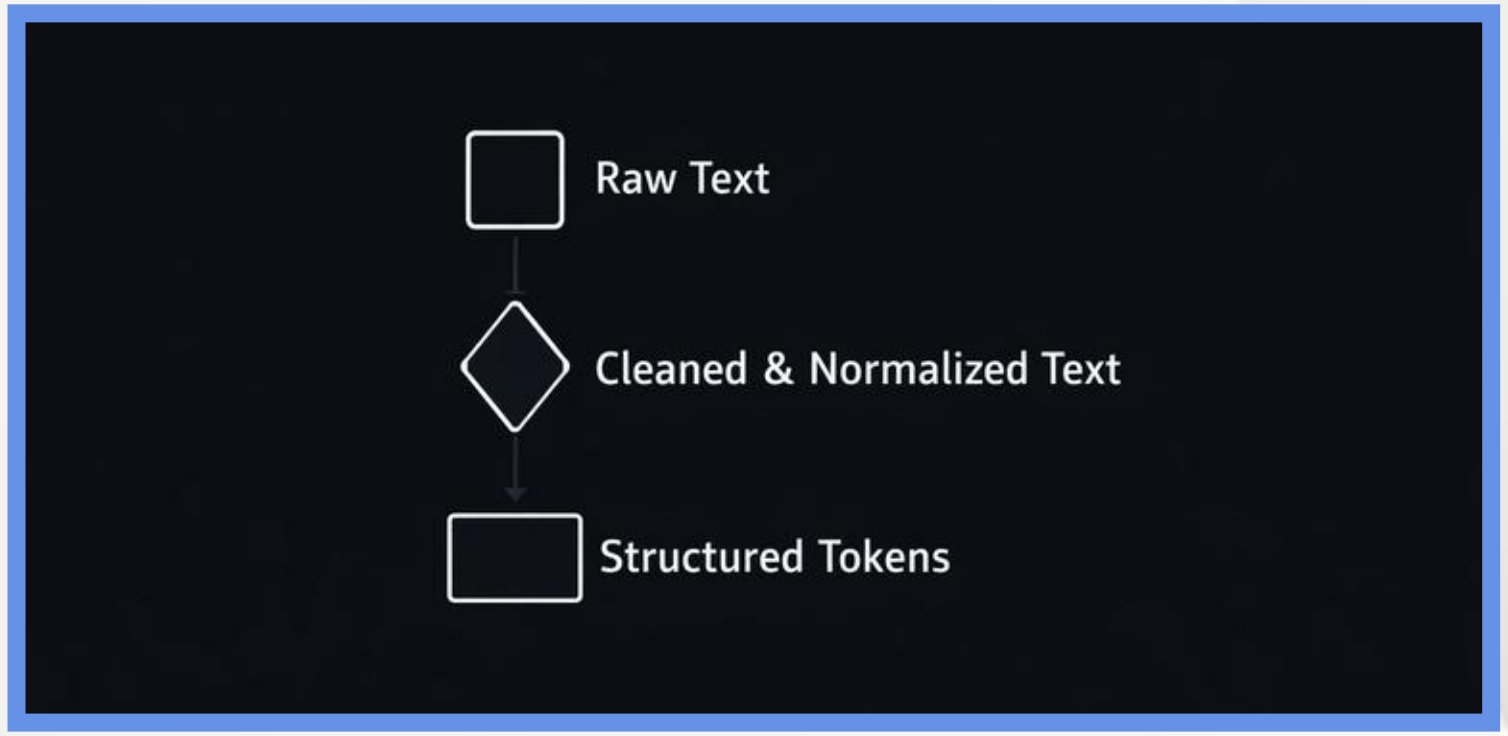
- The goal of preprocessing is to convert raw text into structured, machine-readable units.
- While preserving as much semantic and structural information as possible.
- This step determines the quality and limits of the entire model.



# M01.1 BPE

## Chapter 1.1 Getting Started with Text Processing.

- Cleaning and normalization of noisy text.
- Definition of tokens and vocabulary.
- Preparation for numerical embedding and vectorization.

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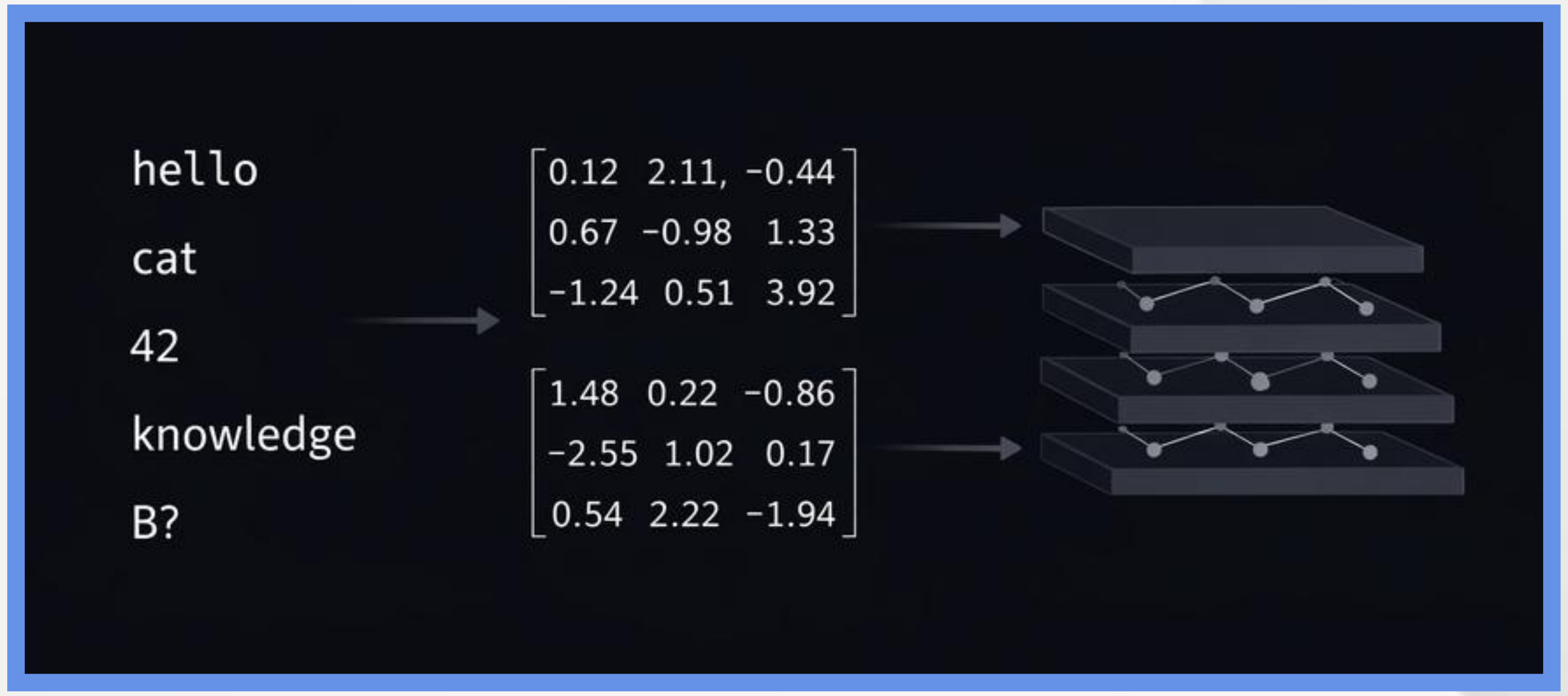
**M01.1 BPE**

# Chapter 1.1 Getting Started with Text Processing.

- Neural networks do not understand symbols or words.
- They operate exclusively on numerical tensors.
- Preprocessing defines how language is translated into numbers, and therefore defines the model's perception of reality.

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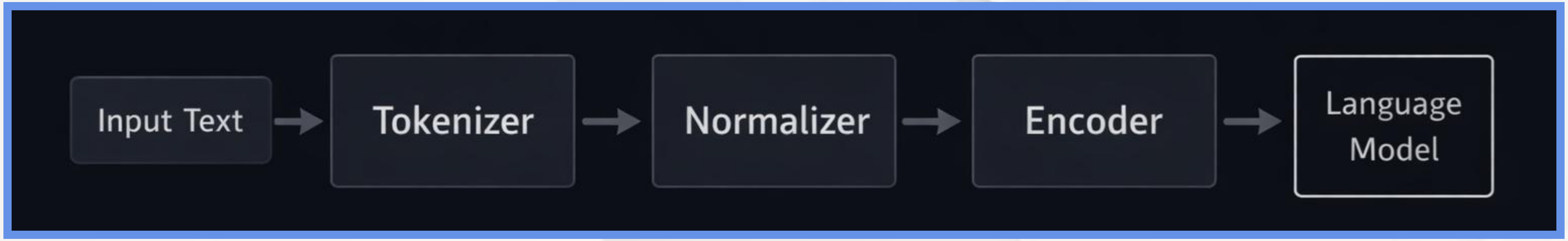
# M01.1 BPE

## Chapter 1.2 Fundamentals of Tokenization

- Modern LLM pipelines rely on highly optimized libraries to ensure efficiency, consistency, and reproducibility.
- These tools define how text becomes training-ready data.

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# Chapter 1.2 Fundamentals of Tokenization

- Tokenization is the deterministic process of splitting text into discrete units called tokens.
- These tokens are the atomic elements used by language models to represent and process human language.


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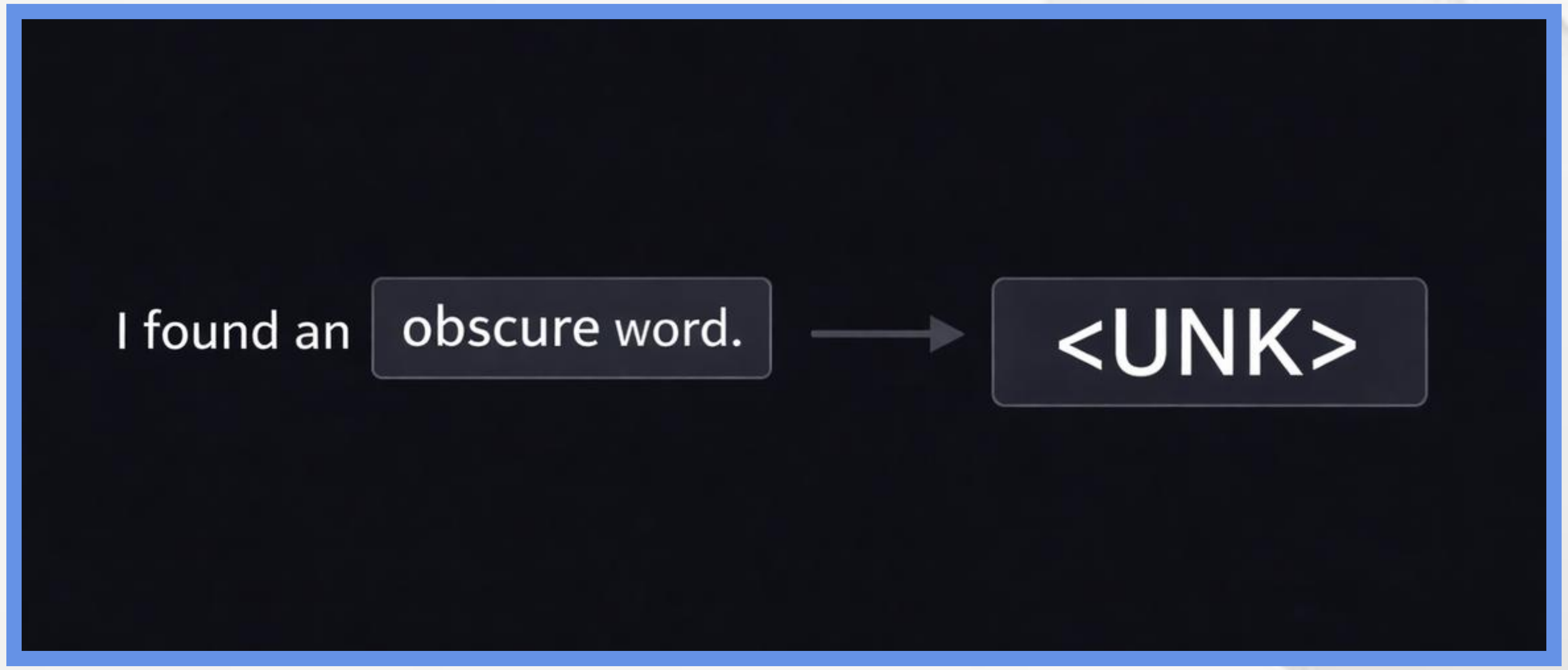
# Chapter 1.2 Fundamentals of Tokenization

- Word-level tokenization fails when encountering words that are not present in the training vocabulary.
- These unknown tokens reduce model performance and limit generalization to new data.

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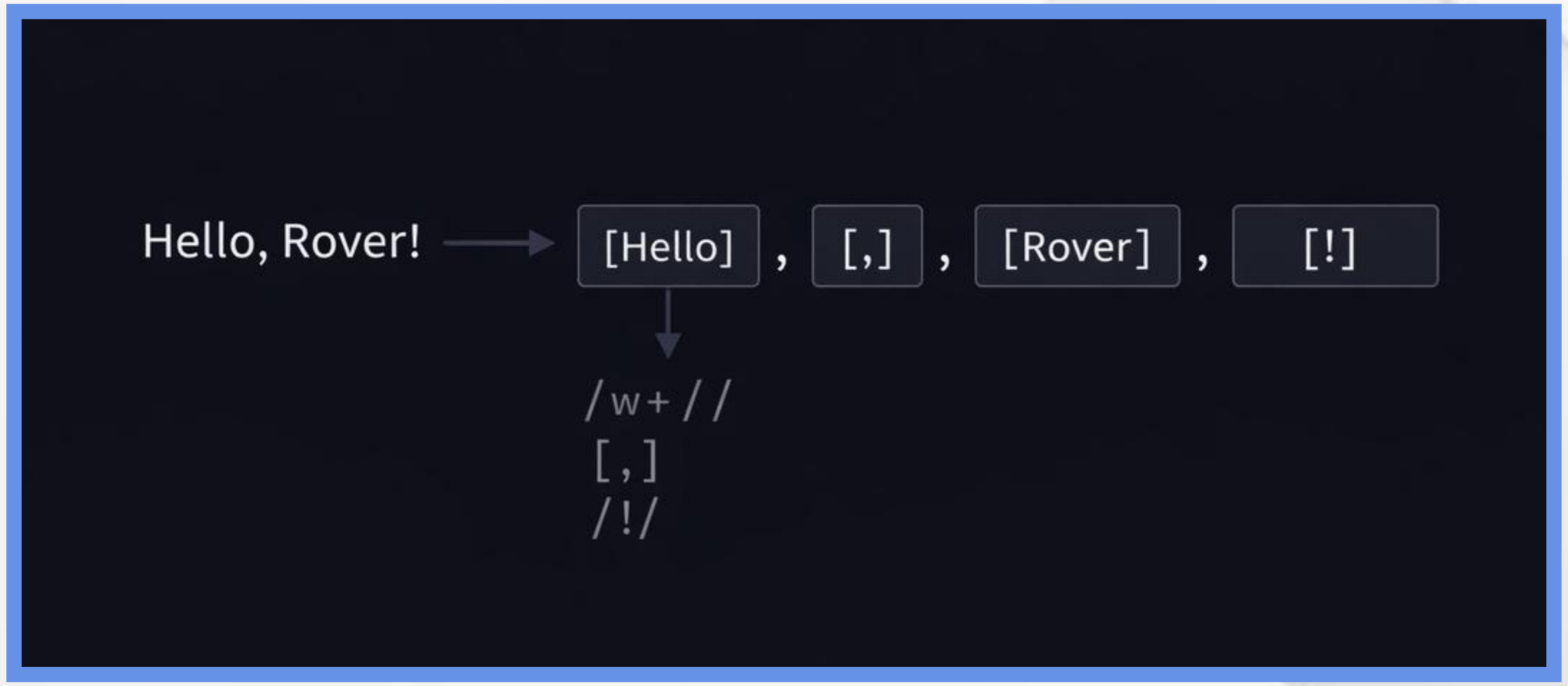
# Chapter 1.2 Fundamentals of Tokenization

- Rule-based tokenizers use predefined patterns to split text while preserving punctuation and structure.
- Regular expressions allow precise control over how symbols, spaces, and contractions are handled.



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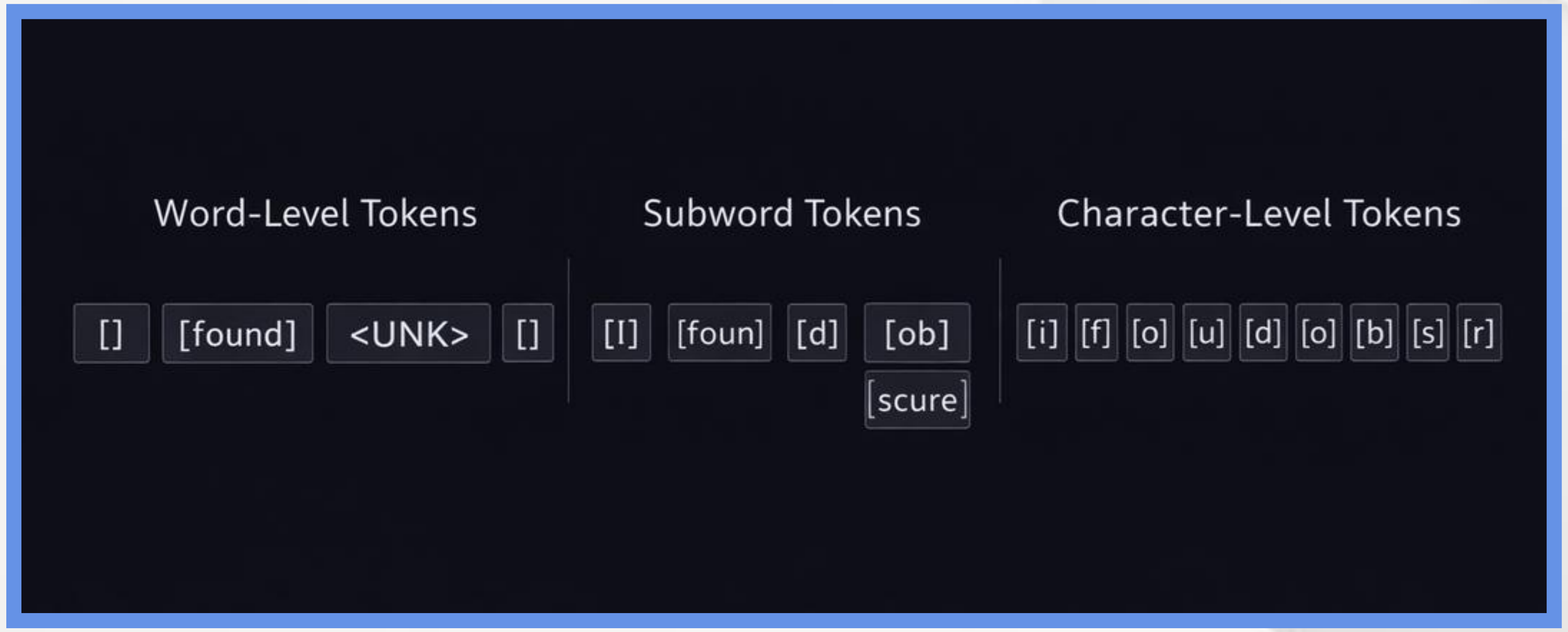
# Chapter 1.3 Byte-Pair Encoding (BPE) and Subword Tokenization

- Subword tokenization was introduced to balance the limitations of word-level and character-level approaches.
- It enables models to represent any word while preserving meaningful linguistic units.

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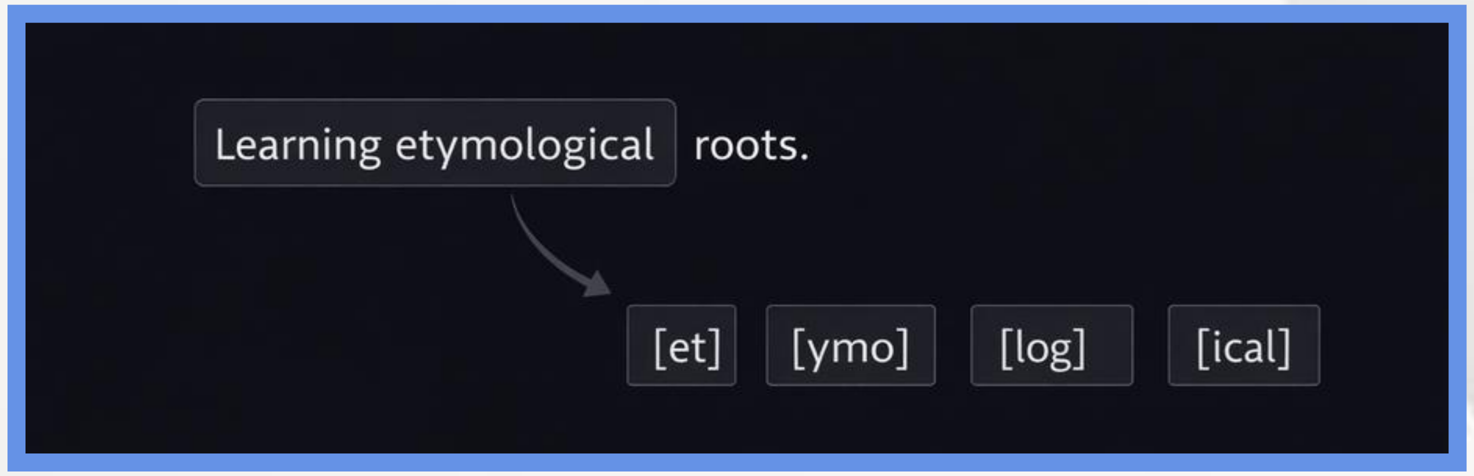
# Chapter 1.3 Byte-Pair Encoding (BPE) and Subword Tokenization

## Handling the Out-of-Vocabulary (OOV) Problem

- Word-level tokenization fails on unseen or rare words.
- Unknown tokens reduce model accuracy and robustness.
- Subword tokenization decomposes unknown words into known units.
- This guarantees full coverage of any input text.

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# Chapter 1.3 Byte-Pair Encoding (BPE) and Subword Tokenization

## What Is Byte-Pair Encoding (BPE)?

- Byte-Pair Encoding is an iterative subword tokenization algorithm.
- It is based on frequency statistics from the training corpus.
- The algorithm merges the most frequent adjacent token pairs.
- Over time, meaningful subword units are learned.


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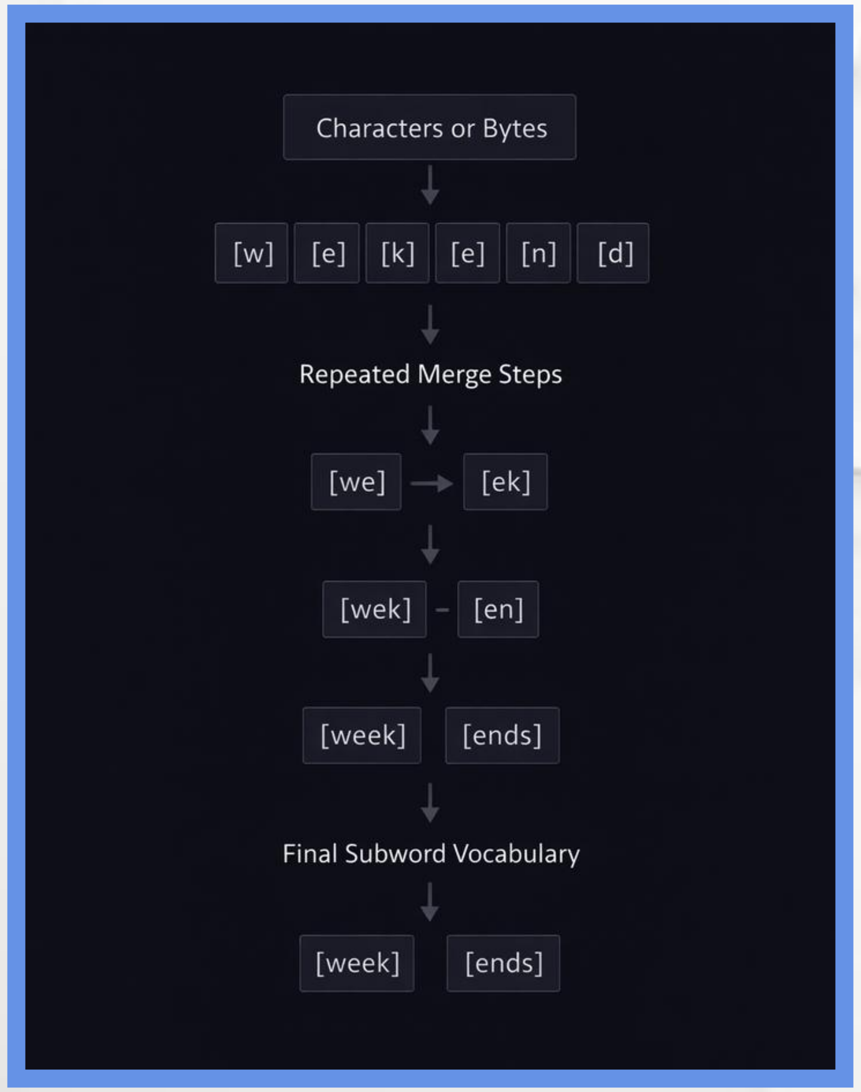
# Chapter 1.3 Byte-Pair Encoding (BPE) and Subword Tokenization

## The BPE Training Process

- Training starts from individual characters or bytes
- All adjacent token pairs are counted across the dataset
- The most frequent pair is merged into a new token
- The process repeats until the target vocabulary size is reached

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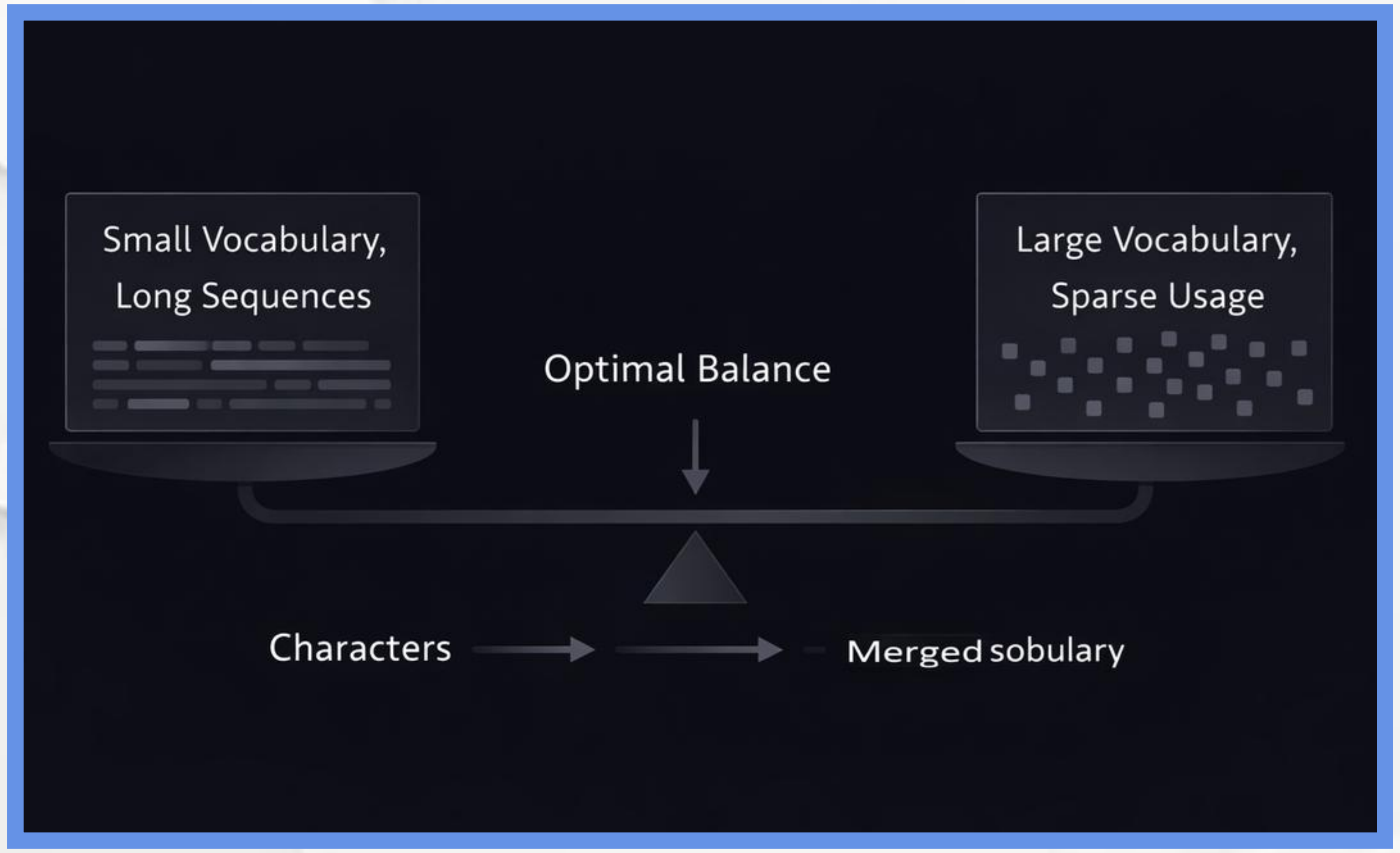
# Chapter 1.3 Byte-Pair Encoding (BPE) and Subword Tokenization

## Vocabulary Size Trade-Offs in BPE

- Small vocabularies increase sequence length
- Longer sequences raise computational cost
- Large vocabularies reduce fragmentation
- Excessively large vocabularies introduce sparsity
- Vocabulary size is a key design decision

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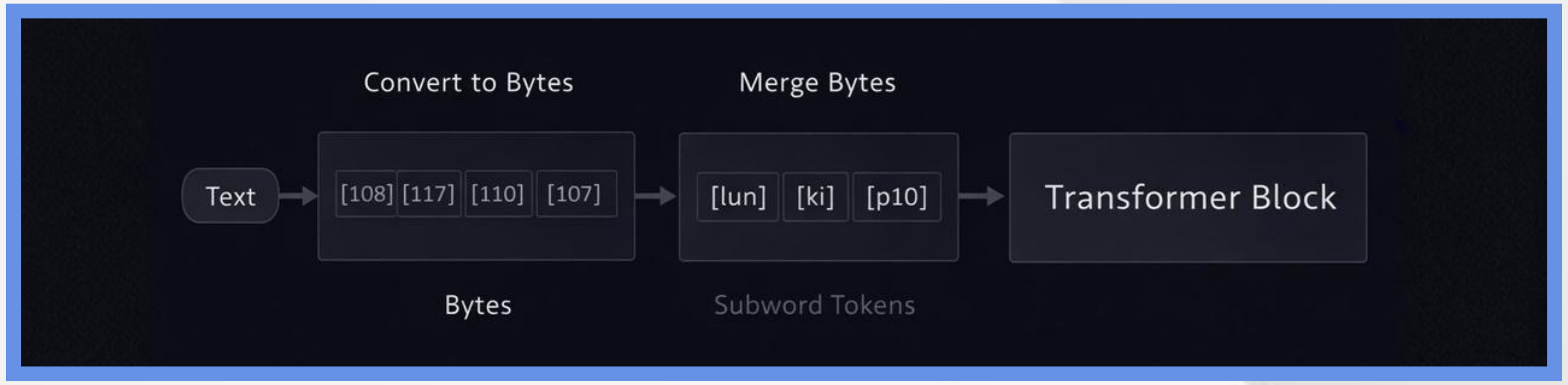
# Chapter 1.3 Byte-Pair Encoding (BPE) and Subword Tokenization

## BPE in Transformer and GPT-Style Models

- Modern Transformer models rely on subword tokenization.
- Byte-level BPE ensures a fixed and complete base vocabulary.
- This approach handles any character or symbol.
- It enables efficient and scalable language modeling.

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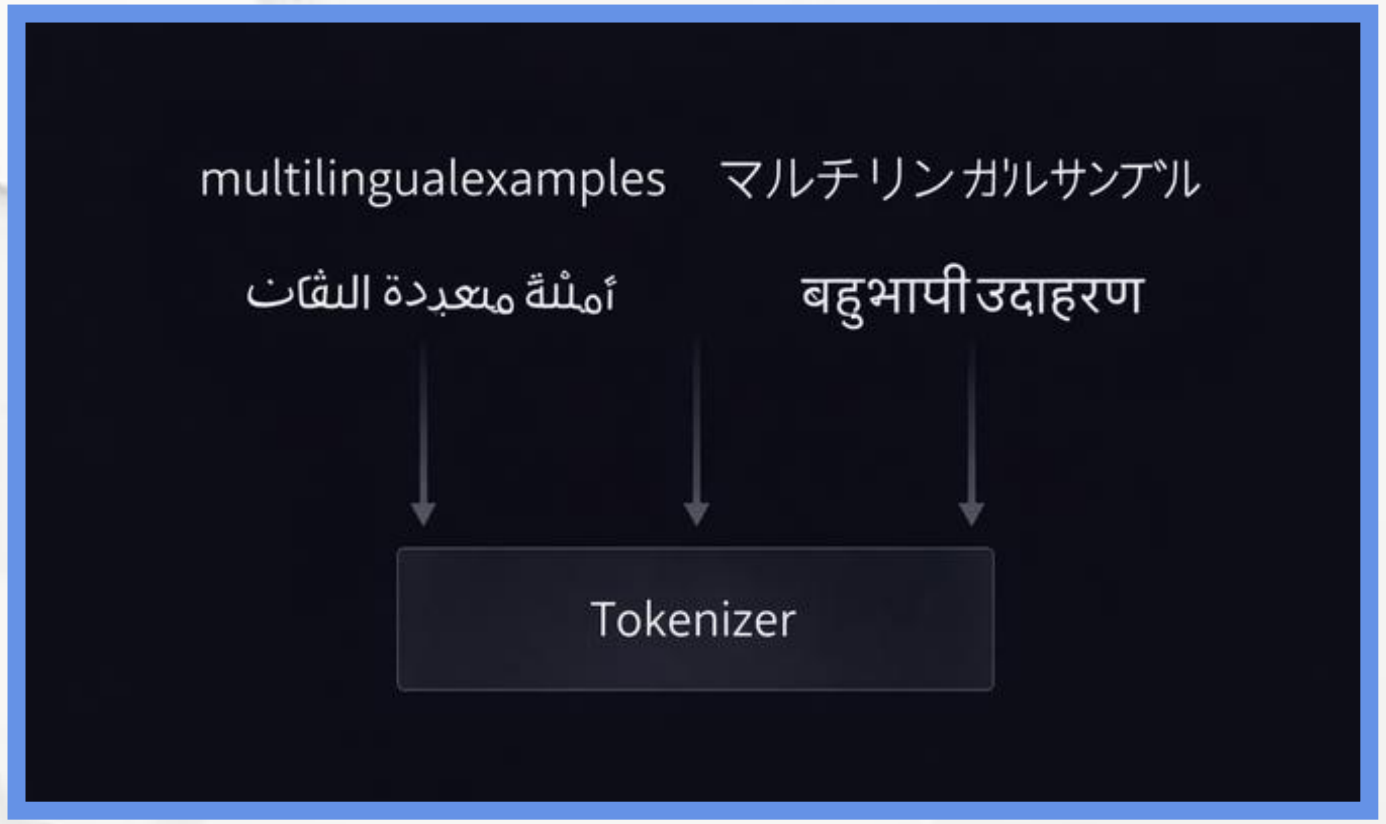
# Chapter 1.4 SentencePiece

## Why SentencePiece Was Introduced

- Traditional tokenizers rely on language-specific rules.
- Whitespace-based pre-tokenization fails for many scripts.
- SentencePiece processes text as a raw character stream.
- This enables fully language-agnostic tokenization.

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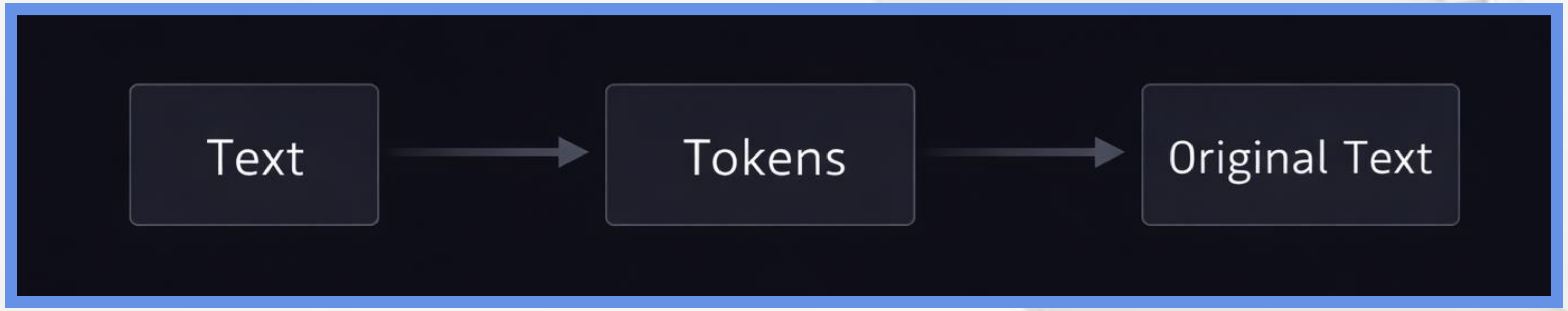
# Chapter 1.5 Stopwords and Vocabulary Management

## Stopwords in Modern NLP Systems

- Stopwords are high-frequency functional words.
- Removing them was common in early NLP systems.
- Transformer models rely on context and attention.
- Modern pipelines often preserve stopwords.

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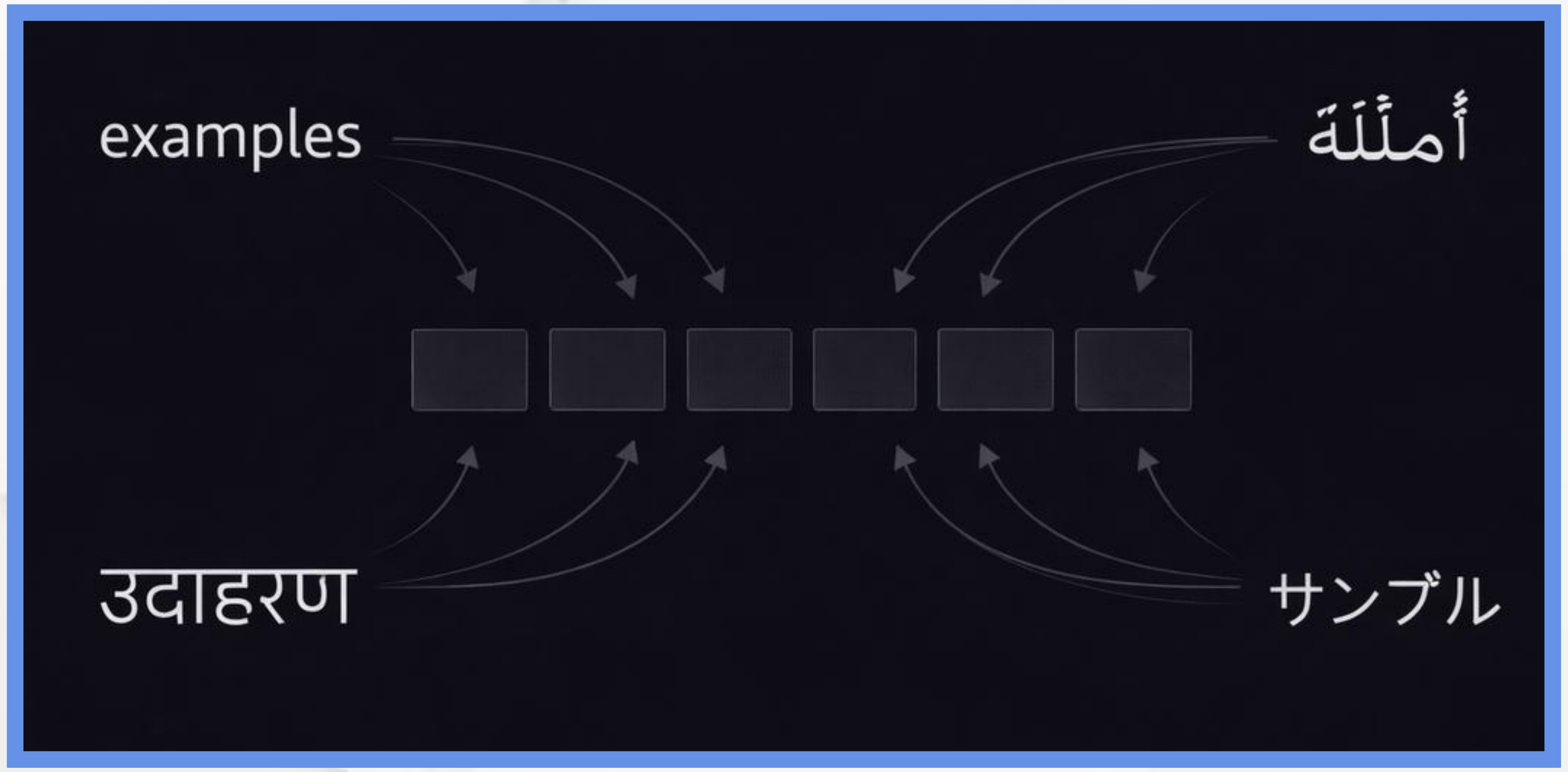
# Chapter 1.6 Processing Multilingual and Cross-Lingual Text

## Challenges of Multilingual Tokenization

- Languages differ in scripts and morphology.
- Whitespace is not universal across languages.
- Subword tokenization enables shared vocabularies.
- This supports cross-lingual and zero-shot learning.

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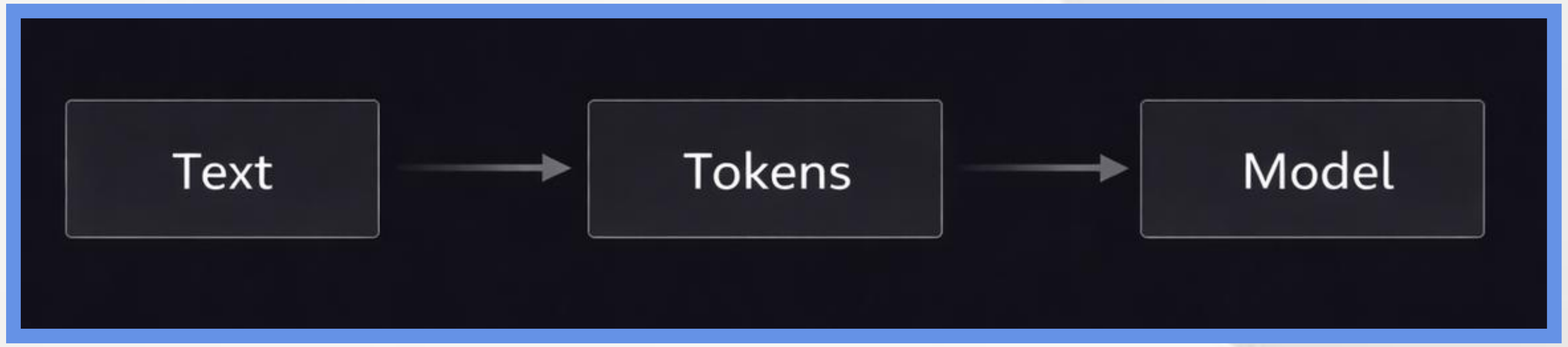
# Chapter 1.7 Summary and Next Steps

## Key Takeaways from Text Preprocessing

- Text preprocessing defines model capabilities.
- Tokenization bridges language and mathematics.
- Subword methods balance flexibility and efficiency.
- Modern systems support multilingual inputs.



# Chapter 1.7 Summary and Next Steps

## Recommended References and Learning Resources

- Sebastian Raschka — Build a Large Language Model (From Scratch)
- Sennrich et al. (2016) — Neural Machine Translation of Rare Words with Subword Units
- Kudo and Richardson (2018) — SentencePiece: A Language-Independent Tokenizer
- Radford et al. (2019) — Language Models are Unsupervised Multitask Learners
- Hugging Face Course — Tokenization and NLP Fundamentals
- OpenAI Documentation — Tokenizers and tiktoken

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